



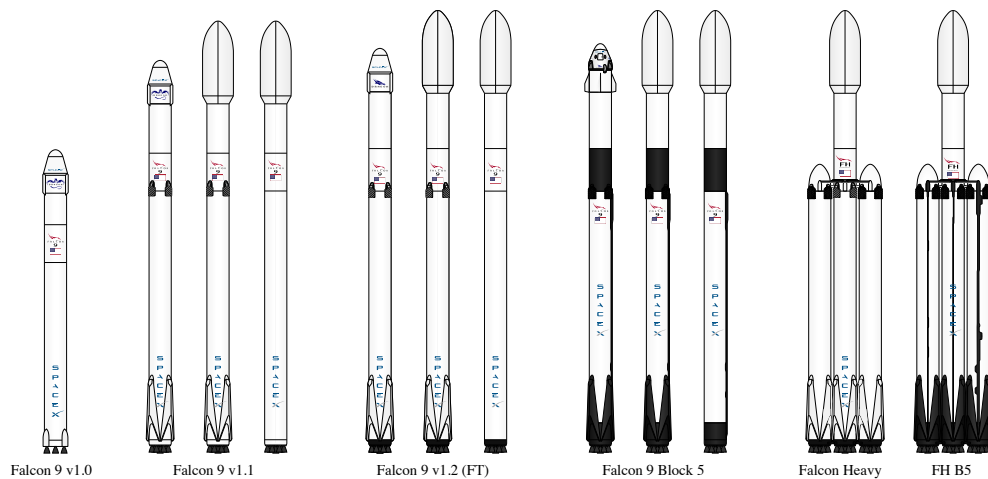
Space X Falcon 9 First Stage Landing Prediction

Web scraping Falcon 9 and Falcon Heavy Launches Records from Wikipedia

Estimated time needed: **40** minutes

In this lab, you will be performing web scraping to collect Falcon 9 historical launch records from a Wikipedia page titled `List of Falcon 9 and Falcon Heavy launches`

https://en.wikipedia.org/wiki/List_of_Falcon_9_and_Falcon_Heavy_launches



Falcon 9 first stage will land successfully



Several examples of an unsuccessful landing are shown here:



More specifically, the launch records are stored in a HTML table shown below:

2020 [\[edit \]](#)

In late 2019, [Gwynne Shotwell](#) stated that SpaceX hoped for as many as 24 launches for Starlink satellites in 2020,^[490] in addition to 14 or 15 non-Starlink launches. At 26 launches, 13 of which for Starlink satellites, Falcon 9 had its most prolific year, and Falcon rockets were second most prolific rocket family of 2020, only behind China's [Long March](#) rocket family.^[491]

[hide] Flight No.	Date and time (UTC)	Version, Booster ^[a]	Launch site	Payload ^[c]	Payload mass	Orbit	Customer	Launch outcome	Booster landing
78	7 January 2020, 02:19:21 ^[492]	F9 B5 Δ B1049.4	CCAFS, SLC-40	Starlink 2 v1.0 (60 satellites)	15,600 kg (34,400 lb) ^[5]	LEO	SpaceX	Success	Success (drone ship)
Third large batch and second operational flight of Starlink constellation. One of the 60 satellites included a test coating to make the satellite less reflective, and thus less likely to interfere with ground-based astronomical observations. ^[493]									
79	19 January 2020, 15:30 ^[494]	F9 B5 Δ B1046.4	KSC, LC-39A	Crew Dragon in-flight abort test ^[495] (Dragon C205.1)	12,050 kg (26,570 lb)	Sub-orbital ^[496]	NASA (CTS) ^[497]	Success	No attempt
An atmospheric test of the Dragon 2 abort system after Max Q . The capsule fired its SuperDraco engines, reached an apogee of 40 km (25 mi), deployed parachutes after reentry, and splashed down in the ocean 31 km (19 mi) downrange from the launch site. The test was previously slated to be accomplished with the Crew Dragon Demo-1 capsule, ^[498] but that test article exploded during a ground test of SuperDraco engines on 20 April 2019. ^[419] The abort test used the capsule originally intended for the first crewed flight. ^[499] As expected, the booster was destroyed by aerodynamic forces after the capsule aborted. ^[500] First flight of a Falcon 9 with only one functional stage — the second stage had a mass simulator in place of its engine.									
80	29 January 2020, 14:07 ^[501]	F9 B5 Δ B1051.3	CCAFS, SLC-40	Starlink 3 v1.0 (60 satellites)	15,600 kg (34,400 lb) ^[5]	LEO	SpaceX	Success	Success
Third operational and fourth large batch of Starlink satellites, deployed in a circular 290 km (180 mi) orbit. One of the fairing halves was caught, while the other was fished out of the ocean. ^[502]									
81	17 February 2020, 15:05 ^[503]	F9 B5 Δ B1056.4	CCAFS, SLC-40	Starlink 4 v1.0 (60 satellites)	15,600 kg (34,400 lb) ^[5]	LEO	SpaceX	Success	Failure (drone ship)
Fourth operational and fifth large batch of Starlink satellites. Used a new flight profile which deployed into a 212 km × 386 km (132 mi × 240 mi) elliptical orbit instead of launching into a circular orbit and firing the second stage engine twice. The first stage booster failed to land on the drone ship ^[504] due to incorrect wind data. ^[505] This was the first time a flight proven booster failed to land.									
82	7 March 2020, 04:50 ^[506]	F9 B5 Δ B1059.2	CCAFS, SLC-40	SpaceX CRS-20 (Dragon C112.3 Δ)	1,977 kg (4,359 lb) ^[507]	LEO (ISS)	NASA (CRS)	Success	Success (ground pad)
Last launch of phase 1 of the CRS contract. Carries Bartolomeo , an ESA platform for hosting external payloads onto ISS. ^[508] Originally scheduled to launch on 2 March 2020, the launch date was pushed back due to a second stage engine failure. SpaceX decided to swap out the second stage instead of replacing the faulty part. ^[509] It was SpaceX's 50th successful landing of a first stage booster, the third flight of the Dragon C112 and the last launch of the cargo Dragon spacecraft.									
83	18 March 2020, 12:16 ^[510]	F9 B5 Δ B1048.5	KSC, LC-39A	Starlink 5 v1.0 (60 satellites)	15,600 kg (34,400 lb) ^[5]	LEO	SpaceX	Success	Failure (drone ship)
Fifth operational launch of Starlink satellites. It was the first time a first stage booster flew for a fifth time and the second time the fairings were reused (Starlink flight in May 2019). ^[511] Towards the end of the first stage burn, the booster suffered premature shut down of an engine, the first of a Merlin 1D variant and first since the CRS-1 mission in October 2012. However, the payload still reached the targeted orbit. ^[512] This was the second Starlink launch booster landing failure in a row, later revealed to be caused by residual cleaning fluid trapped inside a sensor. ^[513]									
84	22 April 2020, 19:30 ^[514]	F9 B5 Δ B1051.4	KSC, LC-39A	Starlink 6 v1.0 (60 satellites)	15,600 kg (34,400 lb) ^[5]	LEO	SpaceX	Success	Success (drone ship)

Objectives

Web scrap Falcon 9 launch records with **BeautifulSoup** :

- Extract a Falcon 9 launch records HTML table from Wikipedia
- Parse the table and convert it into a Pandas data frame

First let's import required packages for this lab

```
In [67]: !pip3 install beautifulsoup4
!pip3 install requests
```

Requirement already satisfied: beautifulsoup4 in /home/jupyterlab/conda/envs/python/lib/python3.7/site-packages (4.11.1)

Requirement already satisfied: soupsieve>1.2 in /home/jupyterlab/conda/envs/python/lib/python3.7/site-packages (from beautifulsoup4) (2.3.2.post1)

Requirement already satisfied: requests in /home/jupyterlab/conda/envs/python/lib/python3.7/site-packages (2.29.0)

Requirement already satisfied: charset-normalizer<4,>=2 in /home/jupyterlab/conda/envs/python/lib/python3.7/site-packages (from requests) (3.1.0)

Requirement already satisfied: idna<4,>=2.5 in /home/jupyterlab/conda/envs/python/lib/python3.7/site-packages (from requests) (3.4)

Requirement already satisfied: urllib3<1.27,>=1.21.1 in /home/jupyterlab/conda/envs/python/lib/python3.7/site-packages (from requests) (1.26.15)

Requirement already satisfied: certifi>=2017.4.17 in /home/jupyterlab/conda/envs/python/lib/python3.7/site-packages (from requests) (2023.5.7)

```
In [68]: import sys

import requests
from bs4 import BeautifulSoup
import re
```

```
import unicodedata
import pandas as pd
```

and we will provide some helper functions for you to process web scraped HTML table

```
In [69]: def date_time(table_cells):
        """
        This function returns the data and time from the HTML table cell
        Input: the element of a table data cell extracts extra row
        """
        return [data_time.strip() for data_time in list(table_cells.strings)][0:2]

    def booster_version(table_cells):
        """
        This function returns the booster version from the HTML table cell
        Input: the element of a table data cell extracts extra row
        """
        out=''.join([booster_version for i,booster_version in enumerate(table_cells.strings) if i%
        return out

    def landing_status(table_cells):
        """
        This function returns the landing status from the HTML table cell
        Input: the element of a table data cell extracts extra row
        """
        out=[i for i in table_cells.strings][0]
        return out

    def get_mass(table_cells):
        mass=unicodedata.normalize("NFKD", table_cells.text).strip()
        if mass:
            mass.find("kg")
            new_mass=mass[0:mass.find("kg")+2]
        else:
            new_mass=0
        return new_mass

    def extract_column_from_header(row):
        """
        This function returns the landing status from the HTML table cell
        Input: the element of a table data cell extracts extra row
        """
        if (row.br):
            row.br.extract()
        if row.a:
            row.a.extract()
        if row.sup:
            row.sup.extract()

        column_name = ' '.join(row.contents)

        # Filter the digit and empty names
        if not(column_name.strip().isdigit()):
            column_name = column_name.strip()
            return column_name
```

To keep the lab tasks consistent, you will be asked to scrape the data from a snapshot of the [List of Falcon 9 and Falcon Heavy launches](#) Wikipedia updated on 9th June 2021

```
In [70]: static_url = "https://en.wikipedia.org/w/index.php?title=List_of_Falcon_9_and_Falcon_Heavy_laun

headers = {
    "User-Agent": "Mozilla/5.0 (Windows NT 10.0; Win64; x64) "
                "AppleWebKit/537.36 (KHTML, like Gecko) "
                "Chrome/91.0.4472.124 Safari/537.36"
}
```

Next, request the HTML page from the above URL and get a `response` object

TASK 1: Request the Falcon9 Launch Wiki page from its URL

First, let's perform an HTTP GET method to request the Falcon9 Launch HTML page, as an HTTP response.

```
In [71]: # use requests.get() method with the provided static_url and headers
# assign the response to a object

response = requests.get(static_url, headers=headers)

# Optional: Check the response
print(f"Status Code: {response.status_code}")
# print(f"Response Content: {response.text}")
```

Status Code: 200

Create a `BeautifulSoup` object from the HTML `response`

```
In [72]: # Use BeautifulSoup() to create a BeautifulSoup object from a response text content

soup = BeautifulSoup(response.text, 'html.parser')

# Optional: Verify the soup object
print(f"Title: {soup.title.string if soup.title else 'No title found'}")
```

Title: List of Falcon 9 and Falcon Heavy launches – Wikipedia

Print the page title to verify if the `BeautifulSoup` object was created properly

```
In [73]: # Use soup.title attribute

title = soup.title

# Print the page title to verify if the BeautifulSoup object was created properly
print(title)
```

<title>List of Falcon 9 and Falcon Heavy launches – Wikipedia</title>

TASK 2: Extract all column/variable names from the HTML table header

Next, we want to collect all relevant column names from the HTML table header

Let's try to find all tables on the wiki page first. If you need to refresh your memory about `BeautifulSoup`, please check the external reference link towards the end of this lab

```
In [74]: # Use the find_all function in the BeautifulSoup object, with element type `table`
# Assign the result to a list called `html_tables`

html_tables = soup.find_all('table')
```

```
In [75]: # html_tables[:1]
html_tables[3]
```

```

Out[75]: <table class="wikitable plainrowheaders collapsible" style="width: 100%;">
<tbody><tr>
<th scope="col">Flight No.
</th>
<th scope="col">Date and<br/>time (<a href="/wiki/Coordinated_Universal_Time" title="Coordinate
d Universal Time">UTC</a>)
</th>
<th scope="col"><a href="/wiki/List_of_Falcon_9_first-stage_boosters" title="List of Falcon 9 f
irst-stage boosters">Version,<br/>Booster</a><sup class="reference" id="cite_ref-booster_11-1">
<a href="#cite_note-booster-11"><span class="cite-bracket">[</span>b<span class="cite-bracke
t">]</span></a></sup>
</th>
<th scope="col">Launch site
</th>
<th scope="col">Payload<sup class="reference" id="cite_ref-Dragon_12-1"><a href="#cite_note-Dra
gon-12"><span class="cite-bracket">[</span>c<span class="cite-bracket">]</span></a></sup>
</th>
<th scope="col">Payload mass
</th>
<th scope="col">Orbit
</th>
<th scope="col">Customer
</th>
<th scope="col">Launch<br/>outcome
</th>
<th scope="col"><a href="/wiki/Falcon_9_first-stage_landing_tests" title="Falcon 9 first-stage
landing tests">Booster<br/>landing</a>
</th></tr>
<tr>
<th rowspan="2" scope="row" style="text-align:center;">8
</th>
<td>6 January 2014,<br/>22:06<sup class="reference" id="cite_ref-NASA_Spaceflight_48-0"><a href
="#cite_note-NASA_Spaceflight-48"><span class="cite-bracket">[</span>41<span class="cite-bracke
t">]</span></a></sup>
</td>
<td><a href="/wiki/Falcon_9_v1.1" title="Falcon 9 v1.1">F9 v1.1</a>
</td>
<td><a href="/wiki/Cape_Canaveral_Space_Force_Station" title="Cape Canaveral Space Force Statio
n">CCAFS</a>,<br/><a href="/wiki/Cape_Canaveral_Space_Launch_Complex_40" title="Cape Canaveral
Space Launch Complex 40">SLC-40</a>
</td>
<td><a href="/wiki/Thaicom_6" title="Thaicom 6">Thaicom 6</a><sup class="reference" id="cite_re
f-sxManifest20120925_28-4"><a href="#cite_note-sxManifest20120925-28"><span class="cite-bracke
t">[</span>22<span class="cite-bracket">]</span></a></sup>
</td>
<td>3,325 kg (7,330 lb)
</td>
<td><a href="/wiki/Geostationary_transfer_orbit" title="Geostationary transfer orbit">GT0</a>
</td>
<td><a href="/wiki/Thaicom" title="Thaicom">Thaicom</a>
</td>
<td class="table-success" style="background: #9EFF9E; color:black; vertical-align: middle; text
-align: center;">Success<sup class="reference" id="cite_ref-sn20140106_49-0"><a href="#cite_not
e-sn20140106-49"><span class="cite-bracket">[</span>42<span class="cite-bracket">]</span></a></sup>
</td>
<td class="table-noAttempt" style="background: #EEE; color:black; vertical-align: middle; text
-align: center;">No attempt<br/><sup class="reference" id="cite_ref-50"><a href="#cite_note-50">
<span class="cite-bracket">[</span>43<span class="cite-bracket">]</span></a></sup>
</td></tr>
<tr>
<td colspan="9">The Thai communication satellite was the second <a href="/wiki/Geostationary_tr
ansfer_orbit" title="Geostationary transfer orbit">GT0</a> launch for Falcon 9. The <a href="/w
iki/United_States_Air_Force" title="United States Air Force">USAF</a> evaluated launch data fro
m this flight as part of a separate certification program for SpaceX to qualify to fly military
payloads, but found that the launch had "unacceptable fuel reserves at engine cutoff of the sta
ge 2 second burnoff".<sup class="reference" id="cite_ref-bloomberg20140722_51-0"><a href="#cite
_note-bloomberg20140722-51"><span class="cite-bracket">[</span>44<span class="cite-bracket">]</span></a></sup>
Thaicom-6 was inserted into a <a href="/wiki/Geostationary_transfer_orbit" titl
e="Geostationary transfer orbit">Super-Synchronous Transfer Orbit</a> of 90,039 km (55,948 mi)
in <a href="/wiki/Apsis" title="Apsis">apogee</a> with an <a href="/wiki/Orbital_inclination" t
itle="Orbital inclination">inclination</a> of 22.46° to the <a href="/wiki/Equator" title="Equa
tor">equator</a>.
</td></tr>
<tr>

```

```

<th rowspan="2" scope="row" style="text-align:center;">9
</th>
<td>18 April 2014,<br/>19:25<sup class="reference" id="cite_ref-SFN_LLog_27-1"><a href="#cite_note-SFN_LLog-27"><span class="cite-bracket">[</span>21<span class="cite-bracket">]</span></a></sup>
</td>
<td><a href="/wiki/Falcon_9_v1.1" title="Falcon 9 v1.1">F9 v1.1</a>
</td>
<td><a class="mw-redirect" href="/wiki/Cape_Canaveral_Air_Force_Station" title="Cape Canaveral Air Force Station">Cape Canaveral</a>,<br/><a href="/wiki/Cape_Canaveral_Space_Launch_Complex_40" title="Cape Canaveral Space Launch Complex 40">LC-40</a>
</td>
<td><a href="/wiki/SpaceX_CRS-3" title="SpaceX CRS-3">SpaceX CRS-3</a><sup class="reference" id="cite_ref-sxManifest20120925_28-5"><a href="#cite_note-sxManifest20120925-28"><span class="cite-bracket">[</span>22<span class="cite-bracket">]</span></a></sup><br/>(Dragon C105)
</td>
<td>2,296 kg (5,062 lb)<sup class="reference" id="cite_ref-52"><a href="#cite_note-52"><span class="cite-bracket">[</span>45<span class="cite-bracket">]</span></a></sup>
</td>
<td><a href="/wiki/Low_Earth_orbit" title="Low Earth orbit">LE0</a> (<a class="mw-redirect" href="/wiki/ISS" title="ISS">ISS</a>)
</td>
<td><a href="/wiki/NASA" title="NASA">NASA</a> (<a href="/wiki/Commercial_Resupply_Services" title="Commercial Resupply Services">CRS</a>)
</td>
<td class="table-success" style="background: #9EFF9E; color:black; vertical-align: middle; text-align: center;">Success
</td>
<td class="partial table-partial" style="background: #BFE; color:black; vertical-align: middle; text-align: center;">Controlled<br/><small>(ocean)</small> <sup class="reference" id="cite_ref-ocean_landing_38-1"><a href="#cite_note-ocean_landing-38"><span class="cite-bracket">[</span>43<span class="cite-bracket">]</span></a></sup><sup class="reference" id="cite_ref-auto_53-0"><a href="#cite_note-auto-53"><span class="cite-bracket">[</span>46<span class="cite-bracket">]</span></a></sup>
</td></tr>
<tr>
<td colspan="9">Following second-stage separation, SpaceX conducted a second <a href="/wiki/Falcon_9_first-stage_landing_tests" title="Falcon 9 first-stage landing tests">controlled-descent test</a> of the discarded booster vehicle and achieved the first successful controlled ocean touchdown of a liquid-rocket-engine orbital booster.<sup class="reference" id="cite_ref-mit20140422_54-0"><a href="#cite_note-mit20140422-54"><span class="cite-bracket">[</span>47<span class="cite-bracket">]</span></a></sup><sup class="reference" id="cite_ref-aw20140428_55-0"><a href="#cite_note-aw20140428-55"><span class="cite-bracket">[</span>48<span class="cite-bracket">]</span></a></sup> Following the soft touchdown, the first stage tipped over as expected and was destroyed. This was the first Falcon 9 booster to fly with extensible landing legs and the first Dragon mission with the <a href="/wiki/Falcon_9_v1.1" title="Falcon 9 v1.1">Falcon 9 v1.1</a> launch vehicle. This flight also launched the <a href="/wiki/Educational_Launch_of_Nanosatellites" title="Educational Launch of Nanosatellites">ELaNa 5</a> mission for <a href="/wiki/NASA" title="NASA">NASA</a> as a secondary payload.<sup class="reference" id="cite_ref-auto2_56-0"><a href="#cite_note-auto2-56"><span class="cite-bracket">[</span>49<span class="cite-bracket">]</span></a></sup><sup class="reference" id="cite_ref-57"><a href="#cite_note-57"><span class="cite-bracket">[</span>50<span class="cite-bracket">]</span></a></sup>
</td></tr>
<tr>
<th rowspan="2" scope="row" style="text-align:center;">10
</th>
<td>14 July 2014,<br/>15:15
</td>
<td><a href="/wiki/Falcon_9_v1.1" title="Falcon 9 v1.1">F9 v1.1</a>
</td>
<td><a class="mw-redirect" href="/wiki/Cape_Canaveral_Air_Force_Station" title="Cape Canaveral Air Force Station">Cape Canaveral</a>,<br/><a href="/wiki/Cape_Canaveral_Space_Launch_Complex_40" title="Cape Canaveral Space Launch Complex 40">LC-40</a>
</td>
<td><a class="mw-redirect" href="/wiki/Orbcomm-0G2" title="Orbcomm-0G2">Orbcomm-0G2</a>-1<br/>(6 satellites)<sup class="reference" id="cite_ref-sxManifest20120925_28-6"><a href="#cite_note-sxManifest20120925-28"><span class="cite-bracket">[</span>22<span class="cite-bracket">]</span></a></sup>
</td>
<td>1,316 kg (2,901 lb)
</td>
<td><a href="/wiki/Low_Earth_orbit" title="Low Earth orbit">LE0</a>
</td>
<td><a href="/wiki/Orbcomm" title="Orbcomm">Orbcomm</a>
</td>

```

```

<td class="table-success" style="background: #9EFF9E; color:black; vertical-align: middle; text-align: center;">Success<sup class="reference" id="cite_ref-og2-01_20140714_58-0"><a href="#cite_note-og2-01_20140714-58"><span class="cite-bracket">[</span>51<span class="cite-bracket">]</span></sup>
</td>
<td class="partial table-partial" style="background: #BFE; color:black; vertical-align: middle; text-align: center;">Controlled<br/><small>(ocean)</small><sup class="reference" id="cite_ref-ocean_landing_38-2"><a href="#cite_note-ocean_landing-38"><span class="cite-bracket">[</span>d<span class="cite-bracket">]</span></sup><sup class="reference" id="cite_ref-auto_53-1"><a href="#cite_note-auto-53"><span class="cite-bracket">[</span>46<span class="cite-bracket">]</span></sup>
</td></tr>
<tr>
<td colspan="9">Payload included six satellites weighing 172 kg (379 lb) each and two 142 kg (313 lb) mass simulators.<sup class="reference" id="cite_ref-gunter-og2_30-1"><a href="#cite_note-gunter-og2-30"><span class="cite-bracket">[</span>24<span class="cite-bracket">]</span></sup><sup class="reference" id="cite_ref-gunter-og2-sim_59-0"><a href="#cite_note-gunter-og2-sim-59"><span class="cite-bracket">[</span>52<span class="cite-bracket">]</span></sup></td>
<td colspan="9">Equipped for the second time with <a class="mw-redirect" href="/wiki/Launch_vehicle_landing_gear" title="Launch vehicle landing gear">landing legs</a>, the first-stage booster successfully conducted a <a class="mw-redirect" href="/wiki/SpaceX_Falcon_9_booster_post-mission,_controlled-descent,_test_program" title="SpaceX Falcon 9 booster post-mission, controlled-descent, test program">controlled-descent</a> test consisting of a burn for deceleration from <a class="mw-redirect" href="/wiki/Hypersonic" title="Hypersonic">hypersonic</a> velocity in the upper atmosphere, a <a href="/wiki/Atmospheric_entry" title="Atmospheric entry">reentry</a> burn, and a final landing burn before soft-landing on the ocean surface.<sup class="reference" id="cite_ref-SpaceX22072014_60-0"><a href="#cite_note-SpaceX22072014-60"><span class="cite-bracket">[</span>53<span class="cite-bracket">]</span></sup>
</td></tr>
<tr>
<th rowspan="2" scope="row" style="text-align:center;">11
</th>
<td>5 August 2014,<br/>08:00
</td>
<td><a href="/wiki/Falcon_9_v1.1" title="Falcon 9 v1.1">F9 v1.1</a>
</td>
<td><a class="mw-redirect" href="/wiki/Cape_Canaveral_Air_Force_Station" title="Cape Canaveral Air Force Station">Cape Canaveral</a>,<br/><a href="/wiki/Cape_Canaveral_Space_Launch_Complex_40" title="Cape Canaveral Space Launch Complex 40">LC-40</a>
</td>
<td><a href="/wiki/AsiaSat_8" title="AsiaSat 8">AsiaSat 8</a><sup class="reference" id="cite_ref-sxManifest20120925_28-7"><a href="#cite_note-sxManifest20120925-28"><span class="cite-bracket">[</span>22<span class="cite-bracket">]</span></sup><sup class="reference" id="cite_ref-AsiaSat_SpaceX_61-0"><a href="#cite_note-AsiaSat_SpaceX-61"><span class="cite-bracket">[</span>54<span class="cite-bracket">]</span></sup><sup class="reference" id="cite_ref-62"><a href="#cite_note-62"><span class="cite-bracket">[</span>55<span class="cite-bracket">]</span></sup>
</td>
<td>4,535 kg (9,998 lb)
</td>
<td><a href="/wiki/Geostationary_transfer_orbit" title="Geostationary transfer orbit">GTO</a>
</td>
<td><a href="/wiki/AsiaSat" title="AsiaSat">AsiaSat</a>
</td>
<td class="table-success" style="background: #9EFF9E; color:black; vertical-align: middle; text-align: center;">Success<sup class="reference" id="cite_ref-as8_20140805_63-0"><a href="#cite_note-as8_20140805-63"><span class="cite-bracket">[</span>56<span class="cite-bracket">]</span></sup>
</td>
<td class="table-noAttempt" style="background: #EEE; color:black; vertical-align: middle; text-align: center;">No attempt<br/><sup class="reference" id="cite_ref-amspace-20140803_64-0"><a href="#cite_note-amspace-20140803-64"><span class="cite-bracket">[</span>57<span class="cite-bracket">]</span></sup>
</td></tr>
<tr>
<td colspan="9">First time SpaceX managed a launch site turnaround between two flights of under a month (22 days). GTO launch of the large communication satellite from Hong Kong did not allow for propulsive return-over-water and controlled splashdown of the first stage.<sup class="reference" id="cite_ref-amspace-20140803_64-1"><a href="#cite_note-amspace-20140803-64"><span class="cite-bracket">[</span>57<span class="cite-bracket">]</span></sup>
</td></tr>
<tr>
<th rowspan="2" scope="row" style="text-align:center;">12
</th>
<td>7 September 2014,<br/>05:00

```

```

</td>
<td><a href="/wiki/Falcon_9_v1.1" title="Falcon 9 v1.1">F9 v1.1</a><br>B1011<sup class="reference" id="cite_ref-block_numbers_14-6"><a href="#cite_note-block_numbers-14"><span class="cite-bracket">[</span>8<span class="cite-bracket">]</span></a></sup>
</td>
<td><a class="mw-redirect" href="/wiki/Cape_Canaveral_Air_Force_Station" title="Cape Canaveral Air Force Station">Cape Canaveral</a>,<br><a href="/wiki/Cape_Canaveral_Space_Launch_Complex_40" title="Cape Canaveral Space Launch Complex 40">LC-40</a>
</td>
<td><a href="/wiki/AsiaSat_6" title="AsiaSat 6">AsiaSat 6</a><sup class="reference" id="cite_ref-sxManifest20120925_28-8"><a href="#cite_note-sxManifest20120925-28"><span class="cite-bracket">[</span>22<span class="cite-bracket">]</span></a></sup><sup class="reference" id="cite_ref-AsiaSat_SpaceX_61-1"><a href="#cite_note-AsiaSat_SpaceX-61"><span class="cite-bracket">[</span>54<span class="cite-bracket">]</span></a></sup><sup class="reference" id="cite_ref-65"><a href="#cite_note-65"><span class="cite-bracket">[</span>58<span class="cite-bracket">]</span></a></sup>
</td>
<td>4,428 kg (9,762 lb)
</td>
<td><a href="/wiki/Geostationary_transfer_orbit" title="Geostationary transfer orbit">GT0</a>
</td>
<td><a href="/wiki/AsiaSat" title="AsiaSat">AsiaSat</a>
</td>
<td class="table-success" style="background: #9EFF9E; color:black; vertical-align: middle; text-align: center;">Success<sup class="reference" id="cite_ref-sdc20140907_66-0"><a href="#cite_note-sdc20140907-66"><span class="cite-bracket">[</span>59<span class="cite-bracket">]</span></a></sup>
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<td class="table-noAttempt" style="background: #EEE; color:black; vertical-align: middle; text-align: center;">No attempt
</td></tr>
<tr>
<td colspan="9">Launch was delayed for two weeks for additional verifications after a malfunction observed in the development of the <a class="mw-redirect" href="/wiki/F9R_Dev1" title="F9R Dev1">F9R Dev1</a> prototype.<sup class="reference" id="cite_ref-67"><a href="#cite_note-67"><span class="cite-bracket">[</span>60<span class="cite-bracket">]</span></a></sup> GT0 launch of the heavy payload did not allow for controlled splashdown.<sup class="reference" id="cite_ref-68"><a href="#cite_note-68"><span class="cite-bracket">[</span>61<span class="cite-bracket">]</span></a></sup>
</td></tr>
<tr>
<th rowspan="2" scope="row" style="text-align:center;">13
</th>
<td>21 September 2014,<br>05:52<sup class="reference" id="cite_ref-SFN_LLog_27-2"><a href="#cite_note-SFN_LLog-27"><span class="cite-bracket">[</span>21<span class="cite-bracket">]</span></a></sup>
</td>
<td><a href="/wiki/Falcon_9_v1.1" title="Falcon 9 v1.1">F9 v1.1</a><br>B1010<sup class="reference" id="cite_ref-block_numbers_14-7"><a href="#cite_note-block_numbers-14"><span class="cite-bracket">[</span>8<span class="cite-bracket">]</span></a></sup>
</td>
<td><a class="mw-redirect" href="/wiki/Cape_Canaveral_Air_Force_Station" title="Cape Canaveral Air Force Station">Cape Canaveral</a>,<br><a href="/wiki/Cape_Canaveral_Space_Launch_Complex_40" title="Cape Canaveral Space Launch Complex 40">LC-40</a>
</td>
<td><a href="/wiki/SpaceX_CRS-4" title="SpaceX CRS-4">SpaceX CRS-4</a><sup class="reference" id="cite_ref-sxManifest20120925_28-9"><a href="#cite_note-sxManifest20120925-28"><span class="cite-bracket">[</span>22<span class="cite-bracket">]</span></a></sup><br>(Dragon <a href="/wiki/Dragon_C106" title="Dragon C106">C106</a>.1)
</td>
<td>2,216 kg (4,885 lb)<sup class="reference" id="cite_ref-69"><a href="#cite_note-69"><span class="cite-bracket">[</span>62<span class="cite-bracket">]</span></a></sup>
</td>
<td><a href="/wiki/Low_Earth_orbit" title="Low Earth orbit">LE0</a> (<a class="mw-redirect" href="/wiki/ISS" title="ISS">ISS</a>)
</td>
<td><a href="/wiki/NASA" title="NASA">NASA</a> (<a href="/wiki/Commercial_Resupply_Services" title="Commercial Resupply Services">CRS</a>)
</td>
<td class="table-success" style="background: #9EFF9E; color:black; vertical-align: middle; text-align: center;">Success<sup class="reference" id="cite_ref-nasacrs420140921_70-0"><a href="#cite_note-nasacrs420140921-70"><span class="cite-bracket">[</span>63<span class="cite-bracket">]</span></a></sup>
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<td class="table-no2" style="background: #FFE3E3; color: black; vertical-align: middle; text-align: center;">

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ign: center;">Uncontrolled<br/><small>(ocean)</small><sup class="reference" id="cite_ref-ocean_
landing_38-3"><a href="#cite_note-ocean_landing-38"><span class="cite-bracket">[</span>d<span c
lass="cite-bracket">]</span></a></sup><sup class="reference" id="cite_ref-fail-13_71-0"><a href
="#cite_note-fail-13-71"><span class="cite-bracket">[</span>64<span class="cite-bracket">]</spa
n></a></sup>
</td></tr>
<tr>
<td colspan="9">Fourth attempt of a soft ocean touchdown,<sup class="reference" id="cite_ref-aw
20141016_72-0"><a href="#cite_note-aw20141016-72"><span class="cite-bracket">[</span>65<span cl
ass="cite-bracket">]</span></a></sup> but the booster ran out of liquid oxygen.<sup class="refe
rence" id="cite_ref-fail-13_71-1"><a href="#cite_note-fail-13-71"><span class="cite-bracket">
[</span>64<span class="cite-bracket">]</span></a></sup> Detailed <a class="mw-redirect" href="/
wiki/Thermal_imaging" title="Thermal imaging">thermal imaging</a> infrared sensor data was coll
ected however by NASA, as part of a joint arrangement with SpaceX as part of research on <a cla
ss="mw-redirect" href="/wiki/Supersonic_retropropulsion" title="Supersonic retropropulsion">ret
ropropulsive deceleration technologies</a> for developing new approaches to Martian <a href="/w
iki/Atmospheric_entry" title="Atmospheric entry">atmospheric entry</a>.<sup class="reference" i
d="cite_ref-aw20141016_72-1"><a href="#cite_note-aw20141016-72"><span class="cite-bracket">[</s
pan>65<span class="cite-bracket">]</span></a></sup>
</td></tr></tbody></table>

```

Starting from the third table is our target table contains the actual launch records.

```

In [76]: # Let's print the third table and check its content
first_launch_table = html_tables[2]
print(first_launch_table)

```

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<table class="wikitable plainrowheaders collapsible" style="width: 100%;">
<tbody><tr>
<th scope="col">Flight No.
</th>
<th scope="col">Date and<br/>time (<a href="/wiki/Coordinated_Universal_Time" title="Coordinated
Universal Time">UTC</a>)
</th>
<th scope="col"><a href="/wiki/List_of_Falcon_9_first-stage_boosters" title="List of Falcon 9 fi
rst-stage boosters">Version,<br/>Booster</a> <sup class="reference" id="cite_ref-booster_11-0"><
a href="#cite_note-booster-11"><span class="cite-bracket">[</span>b<span class="cite-bracket">]<
</span></a></sup>
</th>
<th scope="col">Launch site
</th>
<th scope="col">Payload<sup class="reference" id="cite_ref-Dragon_12-0"><a href="#cite_note-Drag
on-12"><span class="cite-bracket">[</span>c<span class="cite-bracket">]</span></a></sup>
</th>
<th scope="col">Payload mass
</th>
<th scope="col">Orbit
</th>
<th scope="col">Customer
</th>
<th scope="col">Launch<br/>outcome
</th>
<th scope="col"><a href="/wiki/Falcon_9_first-stage_landing_tests" title="Falcon 9 first-stage l
anding tests">Booster<br/>landing</a>
</th></tr>
<tr>
<th rowspan="2" scope="row" style="text-align:center;">1
</th>
<td>4 June 2010,<br/>18:45
</td>
<td><a href="/wiki/Falcon_9_v1.0" title="Falcon 9 v1.0">F9 v1.0</a><sup class="reference" id="ci
te_ref-MuskMay2012_13-0"><a href="#cite_note-MuskMay2012-13"><span class="cite-bracket">[</span>
7<span class="cite-bracket">]</span></a></sup><br/>B0003.1<sup class="reference" id="cite_ref-bl
ock_numbers_14-0"><a href="#cite_note-block_numbers-14"><span class="cite-bracket">[</span>8<spa
n class="cite-bracket">]</span></a></sup>
</td>
<td><a href="/wiki/Cape_Canaveral_Space_Force_Station" title="Cape Canaveral Space Force Statio
n">CCAFS</a>,<br/><a href="/wiki/Cape_Canaveral_Space_Launch_Complex_40" title="Cape Canaveral S
pace Launch Complex 40">SLC-40</a>
</td>
<td><a href="/wiki/Dragon_Spacecraft_Qualification_Unit" title="Dragon Spacecraft Qualification
Unit">Dragon Spacecraft Qualification Unit</a>
</td>
<td>
</td>
<td><a href="/wiki/Low_Earth_orbit" title="Low Earth orbit">LEO</a>
</td>
<td><a href="/wiki/SpaceX" title="SpaceX">SpaceX</a>
</td>
<td class="table-success" style="background: #9EFF9E; color:black; vertical-align: middle; text-
align: center;">Success
</td>
<td class="table-failure" style="background: #FFC7C7; color:black; vertical-align: middle; text-
align: center;">Failure<sup class="reference" id="cite_ref-ns20110930_15-0"><a href="#cite_note-
ns20110930-15"><span class="cite-bracket">[</span>9<span class="cite-bracket">]</span></a></sup>
<sup class="reference" id="cite_ref-16"><a href="#cite_note-16"><span class="cite-bracket">[</span>
10<span class="cite-bracket">]</span></a></sup><br/><small>(parachute)</small>
</td></tr>
<tr>
<td colspan="9">First flight of Falcon 9 v1.0.<sup class="reference" id="cite_ref-sfn20100604_17
-0"><a href="#cite_note-sfn20100604-17"><span class="cite-bracket">[</span>11<span class="cite-b
racket">]</span></a></sup> Used a boilerplate version of Dragon capsule which was not designed t
o separate from the second stage.<small>(<a href="#First_flight_of_Falcon_9">more details below
</a>)</small> Attempted to recover the first stage by parachuting it into the ocean, but it burn
ed up on reentry, before the parachutes even deployed.<sup class="reference" id="cite_ref-parach
ute_18-0"><a href="#cite_note-parachute-18"><span class="cite-bracket">[</span>12<span class="ci
te-bracket">]</span></a></sup>
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<tr>
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<td>8 December 2010,<br/>15:43<sup class="reference" id="cite_ref-spaceflightnow_Clark_Launch_Re

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port_19-0"><a href="#cite_note-spaceflightnow_Clark_Launch_Report-19"><span class="cite-bracket">[</span>13<span class="cite-bracket">]</span></a></sup>
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<td><a href="/wiki/Falcon_9_v1.0" title="Falcon 9 v1.0">F9 v1.0</a><sup class="reference" id="cite_ref-MuskMay2012_13-1"><a href="#cite_note-MuskMay2012-13"><span class="cite-bracket">[</span>7<span class="cite-bracket">]</span></a></sup><br>B0004.1<sup class="reference" id="cite_ref-block_numbers_14-1"><a href="#cite_note-block_numbers-14"><span class="cite-bracket">[</span>8<span class="cite-bracket">]</span></a></sup>
</td>
<td><a href="/wiki/Cape_Canaveral_Space_Force_Station" title="Cape Canaveral Space Force Station">CCAFS</a>,<br><a href="/wiki/Cape_Canaveral_Space_Launch_Complex_40" title="Cape Canaveral Space Launch Complex 40">SLC-40</a>
</td>
<td><a href="/wiki/SpaceX_Dragon" title="SpaceX Dragon">Dragon</a> <a class="mw-redirect" href="/wiki/COTS_Demo_Flight_1" title="COTS Demo Flight 1">demo flight C1</a><br>(Dragon C101)
</td>
<td>
</td>
<td><a href="/wiki/Low_Earth_orbit" title="Low Earth orbit">LEO</a> (<a href="/wiki/International_Space_Station" title="International Space Station">ISS</a>)
</td>
<td><style data-mw-deduplicate="TemplateStyles:r1126788409">.mw-parser-output .plainlist ol,.mw-parser-output .plainlist ul{line-height:inherit;list-style:none;margin:0;padding:0}.mw-parser-output .plainlist ol li,.mw-parser-output .plainlist ul li{margin-bottom:0}</style><div class="plainlist">
<ul><li><a href="/wiki/NASA" title="NASA">NASA</a> (<a href="/wiki/Commercial_Orbital_Transportation_Services" title="Commercial Orbital Transportation Services">COTS</a>)</li>
<li><a href="/wiki/National_Reconnaissance_Office" title="National Reconnaissance Office">NRO</a></li></ul>
</div>
</td>
<td class="table-success" style="background: #9EFF9E; color:black; vertical-align: middle; text-align: center;">Success<sup class="reference" id="cite_ref-ns20110930_15-1"><a href="#cite_note-ns20110930-15"><span class="cite-bracket">[</span>9<span class="cite-bracket">]</span></a></sup>
</td>
<td class="table-failure" style="background: #FFC7C7; color:black; vertical-align: middle; text-align: center;">Failure<sup class="reference" id="cite_ref-ns20110930_15-2"><a href="#cite_note-ns20110930-15"><span class="cite-bracket">[</span>9<span class="cite-bracket">]</span></a></sup>
<sup class="reference" id="cite_ref-20"><a href="#cite_note-20"><span class="cite-bracket">[</span>14<span class="cite-bracket">]</span></a></sup><br><small>(parachute)</small>
</td></tr>
<tr>
<td colspan="9">Maiden flight of <a class="mw-redirect" href="/wiki/Dragon_capsule" title="Dragon capsule">Dragon capsule</a>, consisting of over 3 hours of testing thruster maneuvering and reentry.<sup class="reference" id="cite_ref-spaceflightnow_Clark_unleashing_Dragon_21-0"><a href="#cite_note-spaceflightnow_Clark_unleashing_Dragon-21"><span class="cite-bracket">[</span>15<span class="cite-bracket">]</span></a></sup> Attempted to recover the first stage by parachuting it into the ocean, but it disintegrated upon reentry, before the parachutes were deployed.<sup class="reference" id="cite_ref-parachute_18-1"><a href="#cite_note-parachute-18"><span class="cite-bracket">[</span>12<span class="cite-bracket">]</span></a></sup> <small>(<a href="#COTS_demo_missions">more details below</a>)</small> It also included two <a href="/wiki/CubeSat" title="CubeSat">CubeSats</a>,<sup class="reference" id="cite_ref-NRO_Taps_Boeing_for_Next_Batch_of_CubeSats_22-0"><a href="#cite_note-NRO_Taps_Boeing_for_Next_Batch_of_CubeSats-22"><span class="cite-bracket">[</span>16<span class="cite-bracket">]</span></a></sup> and a wheel of <a href="/wiki/Brou%C3%A8re" title="Brouère">Brouère</a> cheese.
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<tr>
<th rowspan="2" scope="row" style="text-align:center;">3
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<td>22 May 2012,<br>07:44<sup class="reference" id="cite_ref-BBC_new_era_23-0"><a href="#cite_note-BBC_new_era-23"><span class="cite-bracket">[</span>17<span class="cite-bracket">]</span></a></sup>
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<td><a href="/wiki/Falcon_9_v1.0" title="Falcon 9 v1.0">F9 v1.0</a><sup class="reference" id="cite_ref-MuskMay2012_13-2"><a href="#cite_note-MuskMay2012-13"><span class="cite-bracket">[</span>7<span class="cite-bracket">]</span></a></sup><br>B0005.1<sup class="reference" id="cite_ref-block_numbers_14-2"><a href="#cite_note-block_numbers-14"><span class="cite-bracket">[</span>8<span class="cite-bracket">]</span></a></sup>
</td>
<td><a href="/wiki/Cape_Canaveral_Space_Force_Station" title="Cape Canaveral Space Force Station">CCAFS</a>,<br><a href="/wiki/Cape_Canaveral_Space_Launch_Complex_40" title="Cape Canaveral Space Launch Complex 40">SLC-40</a>
</td>
<td><a href="/wiki/SpaceX_Dragon" title="SpaceX Dragon">Dragon</a> <a class="mw-redirect" href="/wiki/Dragon_C2%2B" title="Dragon C2">demo flight C2</a><sup class="reference" id="cite_ref-

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C2_24-0"><a href="#cite_note-C2-24"><span class="cite-bracket">[</span>18<span class="cite-bracket">]</span></a></sup><br/>(Dragon C102)
</td>
<td>525 kg (1,157 lb)<sup class="reference" id="cite_ref-25"><a href="#cite_note-25"><span class="cite-bracket">[</span>19<span class="cite-bracket">]</span></a></sup>
</td>
<td><a href="/wiki/Low_Earth_orbit" title="Low Earth orbit">LE0</a> (<a href="/wiki/International_Space_Station" title="International Space Station">ISS</a>)
</td>
<td><a href="/wiki/NASA" title="NASA">NASA</a> (<a href="/wiki/Commercial_Orbital_Transportation_Services" title="Commercial Orbital Transportation Services">COTS</a>)
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</td>
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<tr>
<td colspan="9">Dragon spacecraft demonstrated a series of tests before it was allowed to approach the <a href="/wiki/International_Space_Station" title="International Space Station">International Space Station</a>. Two days later, it became the first commercial spacecraft to board the ISS.<sup class="reference" id="cite_ref-BBC_new_era_23-1"><a href="#cite_note-BBC_new_era-23"><span class="cite-bracket">[</span>17<span class="cite-bracket">]</span></a></sup> <small>(<a href="#COTS_demo_missions">more details below</a>)</small>
</td></tr>
<tr>
<th rowspan="3" scope="row" style="text-align:center;">4
<td rowspan="2">8 October 2012,<br/>00:35<sup class="reference" id="cite_ref-SFN_LLog_27-0"><a href="#cite_note-SFN_LLog-27"><span class="cite-bracket">[</span>21<span class="cite-bracket">]</span></a></sup>
<td rowspan="2"><a href="/wiki/Falcon_9_v1.0" title="Falcon 9 v1.0">F9 v1.0</a><sup class="reference" id="cite_ref-MuskMay2012_13-3"><a href="#cite_note-MuskMay2012-13"><span class="cite-bracket">[</span>7<span class="cite-bracket">]</span></a></sup><br/>B0006.1<sup class="reference" id="cite_ref-block_numbers_14-3"><a href="#cite_note-block_numbers-14"><span class="cite-bracket">[</span>8<span class="cite-bracket">]</span></a></sup>
<td rowspan="2"><a href="/wiki/Cape_Canaveral_Space_Force_Station" title="Cape Canaveral Space Force Station">CCAFS</a>,<br/><a href="/wiki/Cape_Canaveral_Space_Launch_Complex_40" title="Cape Canaveral Space Launch Complex 40">SLC-40</a>
<td><a href="/wiki/SpaceX_CRS-1" title="SpaceX CRS-1">SpaceX CRS-1</a><sup class="reference" id="cite_ref-sxManifest20120925_28-0"><a href="#cite_note-sxManifest20120925-28"><span class="cite-bracket">[</span>22<span class="cite-bracket">]</span></a></sup><br/>(Dragon C103)
<td>4,700 kg (10,400 lb)
<td><a href="/wiki/Low_Earth_orbit" title="Low Earth orbit">LE0</a> (<a href="/wiki/International_Space_Station" title="International Space Station">ISS</a>)
<td><a href="/wiki/NASA" title="NASA">NASA</a> (<a href="/wiki/Commercial_Resupply_Services" title="Commercial Resupply Services">CRS</a>)
<td class="table-success" style="background: #9EFF9E; color:black; vertical-align: middle; text-align: center;">Success
<td rowspan="2" style="background:#ecec; text-align:center;"><span class="nowrap">No attempt</span>
<td><a href="/wiki/Orbcomm_(satellite)" title="Orbcomm (satellite)">Orbcomm-0G2</a><sup class="reference" id="cite_ref-Orbcomm_29-0"><a href="#cite_note-Orbcomm-29"><span class="cite-bracket">[</span>23<span class="cite-bracket">]</span></a></sup>
<td>172 kg (379 lb)<sup class="reference" id="cite_ref-gunter-og2_30-0"><a href="#cite_note-gunter-og2-30"><span class="cite-bracket">[</span>24<span class="cite-bracket">]</span></a></sup>
<td><a href="/wiki/Low_Earth_orbit" title="Low Earth orbit">LE0</a>
<td><a href="/wiki/Orbcomm" title="Orbcomm">Orbcomm</a>
<td class="table-partial" style="background: #FFB; color:black; vertical-align: middle; text-align: center;">

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gn: center;">Partial failure<sup class="reference" id="cite_ref-nyt-20121030_31-0"><a href="#cite_note-nyt-20121030-31"><span class="cite-bracket">[</span>25<span class="cite-bracket">]</span></a></sup>
</td></tr>
<tr>
<td colspan="9">CRS-1 was successful, but the <a href="/wiki/Secondary_payload" title="Secondary payload">secondary payload</a> was inserted into an abnormally low orbit and subsequently lost. This was due to one of the nine <a href="/wiki/SpaceX_Merlin" title="SpaceX Merlin">Merlin engines</a> shutting down during the launch, and NASA declining a second reignition, as per <a href="/wiki/International_Space_Station" title="International Space Station">ISS</a> visiting vehicle safety rules, the primary payload owner is contractually allowed to decline a second reignition. NASA stated that this was because SpaceX could not guarantee a high enough likelihood of the second stage completing the second burn successfully which was required to avoid any risk of secondary payload's collision with the ISS.<sup class="reference" id="cite_ref-OrbcommTotalLoss_32-0"><a href="#cite_note-OrbcommTotalLoss-32"><span class="cite-bracket">[</span>26<span class="cite-bracket">]</span></a></sup><sup class="reference" id="cite_ref-sn20121011_33-0"><a href="#cite_note-sn20121011-33"><span class="cite-bracket">[</span>27<span class="cite-bracket">]</span></a></sup><sup class="reference" id="cite_ref-34"><a href="#cite_note-34"><span class="cite-bracket">[</span>28<span class="cite-bracket">]</span></a></sup>
</td></tr>
<tr>
<th rowspan="2" scope="row" style="text-align:center;">5
</th>
<td>1 March 2013,<br/>15:10
</td>
<td><a href="/wiki/Falcon_9_v1.0" title="Falcon 9 v1.0">F9 v1.0</a><sup class="reference" id="cite_ref-MuskMay2012_13-4"><a href="#cite_note-MuskMay2012-13"><span class="cite-bracket">[</span>7<span class="cite-bracket">]</span></a></sup><br/>B0007.1<sup class="reference" id="cite_ref-block_numbers_14-4"><a href="#cite_note-block_numbers-14"><span class="cite-bracket">[</span>8<span class="cite-bracket">]</span></a></sup>
</td>
<td><a href="/wiki/Cape_Canaveral_Space_Force_Station" title="Cape Canaveral Space Force Station">CCAFS</a>,<br/><a href="/wiki/Cape_Canaveral_Space_Launch_Complex_40" title="Cape Canaveral Space Launch Complex 40">SLC-40</a>
</td>
<td><a href="/wiki/SpaceX_CRS-2" title="SpaceX CRS-2">SpaceX CRS-2</a><sup class="reference" id="cite_ref-sxManifest20120925_28-1"><a href="#cite_note-sxManifest20120925-28"><span class="cite-bracket">[</span>22<span class="cite-bracket">]</span></a></sup><br/>(Dragon C104)
</td>
<td>4,877 kg (10,752 lb)
</td>
<td><a href="/wiki/Low_Earth_orbit" title="Low Earth orbit">LEO</a> (<a class="mw-redirect" href="/wiki/ISS" title="ISS">ISS</a>)
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<td><a href="/wiki/NASA" title="NASA">NASA</a> (<a href="/wiki/Commercial_Resupply_Services" title="Commercial Resupply Services">CRS</a>)
</td>
<td class="table-success" style="background: #9EFF9E; color:black; vertical-align: middle; text-align: center;">Success
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</td></tr>
<tr>
<td colspan="9">Last launch of the original Falcon 9 v1.0 <a href="/wiki/Launch_vehicle" title="Launch vehicle">launch vehicle</a>, first use of the unpressurized trunk section of Dragon.<sup class="reference" id="cite_ref-sxf9_20110321_35-0"><a href="#cite_note-sxf9_20110321-35"><span class="cite-bracket">[</span>29<span class="cite-bracket">]</span></a></sup>
</td></tr>
<tr>
<th rowspan="2" scope="row" style="text-align:center;">6
</th>
<td>29 September 2013,<br/>16:00<sup class="reference" id="cite_ref-pa20130930_36-0"><a href="#cite_note-pa20130930-36"><span class="cite-bracket">[</span>30<span class="cite-bracket">]</span></a></sup>
</td>
<td><a href="/wiki/Falcon_9_v1.1" title="Falcon 9 v1.1">F9 v1.1</a><sup class="reference" id="cite_ref-MuskMay2012_13-5"><a href="#cite_note-MuskMay2012-13"><span class="cite-bracket">[</span>7<span class="cite-bracket">]</span></a></sup><br/>B1003<sup class="reference" id="cite_ref-block_numbers_14-5"><a href="#cite_note-block_numbers-14"><span class="cite-bracket">[</span>8<span class="cite-bracket">]</span></a></sup>
</td>
<td><a class="mw-redirect" href="/wiki/Vandenberg_Air_Force_Base" title="Vandenberg Air Force Base">VAFB</a>,<br/><a href="/wiki/Vandenberg_Space_Launch_Complex_4" title="Vandenberg Space Launch Complex 4">SLC-4E</a>

```

```

</td>
<td><a href="/wiki/CASSIOPE" title="CASSIOPE">CASSIOPE</a><sup class="reference" id="cite_ref-sxManifest20120925_28-2"><a href="#cite_note-sxManifest20120925-28"><span class="cite-bracket">[</span><span class="cite-bracket">]</span></a></sup><sup class="reference" id="cite_ref-CASSIOPE_MDA_37-0"><a href="#cite_note-CASSIOPE_MDA-37"><span class="cite-bracket">[</span><span class="cite-bracket">]</span></a></sup>
</td>
<td>500 kg (1,100 lb)
</td>
<td><a href="/wiki/Polar_orbit" title="Polar orbit">Polar orbit</a> <a href="/wiki/Low_Earth_orbit" title="Low Earth orbit">LEO</a>
</td>
<td><a href="/wiki/Maxar_Technologies" title="Maxar Technologies">MDA</a>
</td>
<td class="table-success" style="background: #9EFF9E; color:black; vertical-align: middle; text-align: center;">Success<sup class="reference" id="cite_ref-pa20130930_36-1"><a href="#cite_note-pa20130930-36"><span class="cite-bracket">[</span><span class="cite-bracket">]</span></a></sup>
</td>
<td class="table-no2" style="background: #FFE3E3; color: black; vertical-align: middle; text-align: center;">Uncontrolled<br/><small>(ocean)</small><sup class="reference" id="cite_ref-ocean_landing_38-0"><a href="#cite_note-ocean_landing-38"><span class="cite-bracket">[</span><span class="cite-bracket">]</span></a></sup>
</td></tr>
<tr>
<td colspan="9">First commercial mission with a private customer, first launch from Vandenberg, and demonstration flight of Falcon 9 v1.1 with an improved 13-tonne to LEO capacity.<sup class="reference" id="cite_ref-sxf9_20110321_35-1"><a href="#cite_note-sxf9_20110321-35"><span class="cite-bracket">[</span><span class="cite-bracket">]</span></a></sup> After separation from the second stage carrying Canadian commercial and scientific satellites, the first stage booster performed a controlled reentry,<sup class="reference" id="cite_ref-39"><a href="#cite_note-39"><span class="cite-bracket">[</span><span class="cite-bracket">]</span></a></sup> and an <a href="/wiki/Falcon_9_first-stage_landing_tests" title="Falcon 9 first-stage landing tests">ocean touchdown test</a> for the first time. This provided good test data, even though the booster started rolling as it neared the ocean, leading to the shutdown of the central engine as the roll depleted it of fuel, resulting in a hard impact with the ocean.<sup class="reference" id="cite_ref-pa20130930_36-2"><a href="#cite_note-pa20130930-36"><span class="cite-bracket">[</span><span class="cite-bracket">]</span></a></sup> This was the first known attempt of a rocket engine being lit to perform a supersonic retro propulsion, and allowed SpaceX to enter a public-private partnership with <a href="/wiki/NASA" title="NASA">NASA</a> and its Mars entry, descent, and landing technologies research projects.<sup class="reference" id="cite_ref-40"><a href="#cite_note-40"><span class="cite-bracket">[</span><span class="cite-bracket">]</span></a></sup> <small>(<a href="#Maiden_flight_of_v1.1">more details below</a>)</small>
</td></tr>
<tr>
<th rowspan="2" scope="row" style="text-align:center;">7
</th>
<td>3 December 2013,<br/>22:41<sup class="reference" id="cite_ref-sfn_wwls20130624_41-0"><a href="#cite_note-sfn_wwls20130624-41"><span class="cite-bracket">[</span><span class="cite-bracket">]</span></a></sup>
</td>
<td><a href="/wiki/Falcon_9_v1.1" title="Falcon 9 v1.1">F9 v1.1</a><br/>B1004
</td>
<td><a href="/wiki/Cape_Canaveral_Space_Force_Station" title="Cape Canaveral Space Force Station">CCAFS</a>,<br/><a href="/wiki/Cape_Canaveral_Space_Launch_Complex_40" title="Cape Canaveral Space Launch Complex 40">SLC-40</a>
</td>
<td><a href="/wiki/SES-8" title="SES-8">SES-8</a><sup class="reference" id="cite_ref-sxManifest20120925_28-3"><a href="#cite_note-sxManifest20120925-28"><span class="cite-bracket">[</span><span class="cite-bracket">]</span></a></sup><sup class="reference" id="cite_ref-spx-pr_42-0"><a href="#cite_note-spx-pr-42"><span class="cite-bracket">[</span><span class="cite-bracket">]</span></a></sup><sup class="reference" id="cite_ref-aw20110323_43-0"><a href="#cite_note-aw20110323-43"><span class="cite-bracket">[</span><span class="cite-bracket">]</span></a></sup>
</td>
<td>3,170 kg (6,990 lb)
</td>
<td><a href="/wiki/Geostationary_transfer_orbit" title="Geostationary transfer orbit">GT0</a>
</td>
<td><a class="mw-redirect" href="/wiki/SES_S.A." title="SES S.A.">SES</a>
</td>
<td class="table-success" style="background: #9EFF9E; color:black; vertical-align: middle; text-align: center;">Success<sup class="reference" id="cite_ref-SNMissionStatus7_44-0"><a href="#cite_note-SNMissionStatus7-44"><span class="cite-bracket">[</span><span class="cite-bracket">]</span></a></sup>
</td>

```

```
 No attempt<br/><sup class="reference" id="cite_ref-sf10120131203_45-0"><a href="#cite_note-sf10120131203-45"><span class="cite-bracket">[</span>38<span class="cite-bracket">]</span></a></sup> </td></tr> <tr> <td colspan="9">First <a href="/wiki/Geostationary_transfer_orbit" title="Geostationary transfer orbit">Geostationary transfer orbit</a> (GTO) launch for Falcon 9,<sup class="reference" id="cite_ref-spx-pr_42-1"><a href="#cite_note-spx-pr-42"><span class="cite-bracket">[</span>35<span class="cite-bracket">]</span></a></sup> and first successful reignition of the second stage.<sup class="reference" id="cite_ref-46"><a href="#cite_note-46"><span class="cite-bracket">[</span>39<span class="cite-bracket">]</span></a></sup> SES-8 was inserted into a <a href="/wiki/Geostationary_transfer_orbit" title="Geostationary transfer orbit">Super-Synchronous Transfer Orbit</a> of 79,341 km (49,300 mi) in apogee with an <a href="/wiki/Orbital_inclination" title="Orbital inclination">inclination</a> of 20.55° to the <a href="/wiki/Equator" title="Equator">equator</a>. </td></tr></tbody></table> |
```

You should be able to see the column names embedded in the table header elements `<th>` as follows:

```

<tr>
<th scope="col">Flight No.
</th>
<th scope="col">Date and<br/>time (<a href="/wiki/Coordinated_Universal_Time" title="Coordinated Universal Time">UTC</a>)
</th>
<th scope="col"><a href="/wiki/List_of_Falcon_9_first-stage_boosters" title="List of Falcon 9 first-stage boosters">Version,<br/>Booster</a> <sup class="reference" id="cite_ref-boosters_11-0"><a href="#cite_note-boosters-11">[b]</a></sup>
</th>
<th scope="col">Launch site
</th>
<th scope="col">Payload<sup class="reference" id="cite_ref-Dragon_12-0"><a href="#cite_note-Dragon-12">[c]</a></sup>
</th>
<th scope="col">Payload mass
</th>
<th scope="col">Orbit
</th>
<th scope="col">Customer
</th>
<th scope="col">Launch<br/>outcome
</th>
<th scope="col"><a href="/wiki/Falcon_9_first-stage_landing_tests" title="Falcon 9 first-stage landing tests">Booster<br/>landing</a>
</th></tr>

```

Next, we just need to iterate through the `<th>` elements and apply the provided

`extract_column_from_header()` to extract column name one by one

```

In [77]: column_names = []

# Apply find_all() function with `th` element on first_launch_table
# Iterate each th element and apply the provided extract_column_from_header() to get a column name
# Append the Non-empty column name (if name is not None and len(name) > 0) into a list called

for i in first_launch_table.find_all('th'):
    name = extract_column_from_header(i) # apply function to get a column name
    if name is not None and len(name) > 0:
        column_names.append(name)

```

Check the extracted column names

```

In [78]: print(column_names)

['Flight No.', 'Date and time ( )', 'Launch site', 'Payload', 'Payload mass', 'Orbit', 'Customer', 'Launch outcome']

```

TASK 3: Create a data frame by parsing the launch HTML tables

We will create an empty dictionary with keys from the extracted column names in the previous task. Later, this dictionary will be converted into a Pandas dataframe

```
In [79]: launch_dict= dict.fromkeys(column_names)

# Remove an irrelevant column
del launch_dict['Date and time ( )']

# Let's initial the launch_dict with each value to be an empty list
launch_dict['Flight No.'] = []
launch_dict['Launch site'] = []
launch_dict['Payload'] = []
launch_dict['Payload mass'] = []
launch_dict['Orbit'] = []
launch_dict['Customer'] = []
launch_dict['Launch outcome'] = []
# Added some new columns
launch_dict['Version Booster']=[]
launch_dict['Booster landing']=[]
launch_dict['Date']=[]
launch_dict['Time']=[]
```

Next, we just need to fill up the `launch_dict` with launch records extracted from table rows.

Usually, HTML tables in Wiki pages are likely to contain unexpected annotations and other types of noises, such as reference links `B0004.1[8]` , missing values `N/A [e]` , inconsistent formatting, etc.

To simplify the parsing process, we have provided an incomplete code snippet below to help you to fill up the `launch_dict`. Please complete the following code snippet with TODOs or you can choose to write your own logic to parse all launch tables:

```
In [80]: print("Keys in launch_dict:", list(launch_dict.keys()))
print("Column names:", column_names)
```

```
Keys in launch_dict: ['Flight No.', 'Launch site', 'Payload', 'Payload mass', 'Orbit', 'Customer', 'Launch outcome', 'Version Booster', 'Booster landing', 'Date', 'Time']
Column names: ['Flight No.', 'Date and time ( )', 'Launch site', 'Payload', 'Payload mass', 'Orbit', 'Customer', 'Launch outcome']
```

```
In [87]: extracted_row = 0
#Extract each table
for table_number,table in enumerate(soup.find_all('table',"wikitable plainrowheaders collapsible")):
    # get table row
    for rows in table.find_all("tr"):
        #check to see if first table heading is as number corresponding to launch a number
        if rows.th:
            if rows.th.string:
                flight_number=rows.th.string.strip()
                flag=flight_number.isdigit()
            else:
                flag=False
        #get table element
        row=rows.find_all('td')
        #if it is number save cells in a dictionary
        if flag:
            extracted_row += 1
            # Flight Number value
            # TODO: Append the flight_number into launch_dict with key `Flight No.`
            # launch_dict['Flight No.'] = flight_number
            launch_dict['Flight No.'].append(flight_number)
            #print(flight_number)
            datatimelist=date_time(row[0])

            # Date value
            # TODO: Append the date into launch_dict with key `Date`
            date = datatimelist[0].strip(',')
            # launch_dict['Date'] = date
            launch_dict['Date'].append(date)
            #print(date)

            # Time value
```



```

# TODO: Append the time into launch_dict with key `Time`
time = datatimelist[1]
# launch_dict['Time']=time
launch_dict['Time'].append(time)
#print(time)

# Booster version
# TODO: Append the bv into launch_dict with key `Version Booster`
bv=booster_version(row[1])
if not(bv):
    bv=row[1].a.string
print(bv)
# launch_dict['Version Booster'] = bv
launch_dict['Version Booster'].append(bv)

# Launch Site
# TODO: Append the bv into launch_dict with key `Launch Site`
launch_site = row[2].a.string
#print(launch_site)
# launch_dict['Launch Site'] = launch_site
launch_dict['Launch site'].append(launch_site)

# Payload
# TODO: Append the payload into launch_dict with key `Payload`
payload = row[3].a.string
#print(payload)
# launch_dict['Payload'] = payload
launch_dict['Payload'].append(payload)

# Payload Mass
# TODO: Append the payload_mass into launch_dict with key `Payload mass`
payload_mass = get_mass(row[4])
#print(payload)
# launch_dict['Payload mass'] = payload_mass
launch_dict['Payload mass'].append(payload_mass)

# Orbit
# TODO: Append the orbit into launch_dict with key `Orbit`
orbit = row[5].a.string
#print(orbit)
# launch_dict['Orbit'] = orbit
launch_dict['Orbit'].append(orbit)

# # Customer
# # TODO: Append the customer into launch_dict with key `Customer`
# customer = row[6].a.string
# #print(customer)
# # launch_dict['Customer'] = customer
# launch_dict['Customer'].append(customer)

# Customer
try:
    customer = row[6].a.string
except AttributeError:
    customer = row[6].get_text(strip=True)

#
    launch_dict['Customer'] = customer
launch_dict['Customer'].append(customer)

# Launch outcome
# TODO: Append the launch_outcome into launch_dict with key `Launch outcome`
launch_outcome = list(row[7].strings)[0]
#print(launch_outcome)
# launch_dict['Launch outcome'] = launch_outcome
launch_dict['Launch outcome'].append(launch_outcome)

# Booster landing
# TODO: Append the launch_outcome into launch_dict with key `Booster landing`
booster_landing = landing_status(row[8])
#print(booster_landing)
# launch_dict['Booster landing'] = booster_landing
launch_dict['Booster landing'].append(booster_landing)

```

18/20

```

F9 B5[
F9 B5B1056.3482
F9 B5
F9 B5
F9 B5
F9 B5
F9 B5
F9 B5
F9 B5[
F9 B5
F9 B5
F9 B5
F9 B5B1058.2544
F9 B5
F9 B5B1049.6544
F9 B5
F9 B5B1060.2563
F9 B5B1058.3565
F9 B5B1051.6568
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F9 B5
F9 B5B1051.8609
F9 B5B1058.5613
F9 B5 △[
F9 B5 △
F9 B5 △[
F9 B5 △[
F9 B5 △
F9 B5B1060.6643
F9 B5 △
F9 B5B1061.2647
F9 B5B1060.7652
F9 B5B1049.9655
F9 B5B1051.10657
F9 B5B1058.8660
F9 B5B1063.2665
F9 B5B1067.1668
F9 B5

```

After you have fill in the parsed launch record values into `launch_dict`, you can create a dataframe from it.

```
In [88]: df = pd.DataFrame({ key:pd.Series(value) for key, value in launch_dict.items() })
```

```
In [89]: df
```

Out [89]:

	Flight No.	Launch site	Payload	Payload mass	Orbit	Customer	Launch outcome	Version Booster	Booster landing	Date
0	1	CCAFS	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success\n	F9 v1.07B0003.18	Failure	4 Jul 2010
1	1	CCAFS	Dragon	0	LEO	NASA	Success	F9 v1.07B0003.18	Failure	4 Jul 2010
2	2	CCAFS	Dragon	525 kg	LEO	NASA	Success	F9 v1.07B0004.18	No attempt\n	Deceml 2010
3	3	CCAFS	SpaceX CRS-1	4,700 kg	LEO	NASA	Success\n	F9 v1.07B0005.18	No attempt	22 Mar 2012
4	4	CCAFS	SpaceX CRS-2	4,877 kg	LEO	NASA	Success\n	F9 v1.07B0006.18	No attempt\n	8 Octol 2012
...	...	...	...	...	...	...	...	...	...	
223	117	KSC	Starlink	~14,000 kg	LEO	SpaceX	Success\n	F9 B5B1051.10657	Success	9 Mar 2022
224	118	CCSFS	Starlink	15,600 kg	LEO	NASA	Success\n	F9 B5B1058.8660	Success	15 Mar 2022
225	119	KSC	SpaceX CRS-22	3,328 kg	LEO	Sirius XM	Success\n	F9 B5B1063.2665	Success	26 Mar 2022
226	120	CCSFS	SXM-8	7,000 kg	GTO	NaN	NaN	F9 B5B1067.1668	NaN	3 Jul 2022
227	121	NaN	NaN	NaN	NaN	NaN	NaN	F9 B5	NaN	6 Jul 2022

228 rows x 11 columns

We can now export it to a **CSV** for the next section, but to make the answers consistent and in case you have difficulties finishing this lab.

Following labs will be using a provided dataset to make each lab independent.

```
df.to_csv('spacex_web_scraped.csv', index=False)
```

```
In [90]: df.to_csv('spacex_web_scraped.csv', index=False)
```

Authors

[Yan Luo](#)

[Nayef Abou Tayoun](#)

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